

as we saw earlier, may spoof IP, MAC, and SSID. Kismet can fight back against these techniques because it supports SSID declocking and MAC spoofing.

Kismet is a fully capable and complete tool for detecting rogue APs on a network. It can be installed on a laptop, PC or a hand-held device with a wireless NIC (network interface card) to scan wireless packet data transfers. The tool comes with an added advantage. It can simultaneously access more than one NIC to collect packets. Such multiplexing is very helpful for collecting data at a rapid rate. As explained earlier, all the frequencies cannot be scanned simultaneously. However, by using multiple NICs one can speed up the process. Current versions of Kismet are limited to detecting only 802.11b wireless devices, and if you are in luck, Kismet might be able to detect an 802.11g device since it's backward-compatible with 802.11b.

As Kismet detects devices, it can predict the location of the rogue AP and plot it on a map if a GPS card is connected to the system. But this is experimental, as large metal objects and radio frequency interference attenuate the signal, making detection difficult. As wireless devices are being detected, the information to identify the node is recorded. For example, the SSID, the type of the device, manufacturer, etc. is recorded.

To actually find the location of the rogue AP, you must walk around in the network area with a wireless device that has Kismet, and let the tool detect any intrusion. Currently, having to move with the device is its only drawback. As you move closer and closer to the rogue point, the signal strength goes on increasing and the location can be pointed out.

Kismet is available as a free download from kismetwireless.net.

Tracking other types of intrusions using Kismet

Kismet can give detailed information about the types of attacks we have discussed, including DoS attacks or network breaches using tools such as NetStumbler. It also detects probing of networks by clients or nodes that are not part of the network, yet are sniffing the network. Kismet focuses on the 802.11 network standard. It also provides alerts to users when a problematic condition occurs.

Alerts provide a lot of help when someone is monitoring the network. Some are very useful and reliable, but others may generate false or useless information. The following is an example:

```
Alert name: NETSTUMBLER
Alert type: Fingerprint
Alert on: Netstumbler probe requests
WVE: WVE-2005-0025
Alert message: "Netstumbler ($version) probe detected from ($macsource)"
Tool-specific: Yes (Netstumbler 3.22, 3.23, 3.30)
References: http://www.netstumbler.com
Details: In an attempt to disclose the SSID of a network,
Netstumbler sends out unique packets. This is not done
in all situations, but when it is detected the potential
for false positives is very low
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What you can see above is a typical Kismet output (taken from the official Kismet documentation) when it finds some rogue client trying to access information from your network using a NetStumbler packet sniffer tool. To know all the types of intrusions Kismet can detect, please check the documentation at www.kismetwireless.net/documentation.shtml.

Wireless IDS drawbacks

The benefits of a wireless IDS are numerous, but there are a few drawbacks to be considered before deploying such a system in any network. Wireless intrusion detection is an experimental technology and proper evaluation is needed before its application. Because the technology is new, there are possibilities of bugs in the code, loopholes, or worse, vulnerabilities. These could potentially actually weaken the network if you fail to integrate the IDS effectively with your network. Having said that, with the current pace of research in this field, and newer and better algorithms and tools being released every day, most of the vulnerabilities will be gone in no time.

A wireless IDS is only as effective as the individuals who analyse and respond to the data gathered by the system (again, the network is only as strong as its weakest link/node). A wireless IDS, like a standard IDS, can require vast man, machine and money resources to analyse and respond to the collected data. Generally, more manpower is needed to monitor wireless IDS as opposed to standard IDS. The reason is simple! You not only have to analyse the data from various frequencies and bands, but also have to move around trying to find rogue APs. However, this point is debatable.

Given the advantages of these systems, these drawbacks (except the bugs, which could weaken the system) seem minor.

We have considered all the aspects in case of wireless IDS. Along with the strong encryption and the firewalls in networks, the IDS forms a second line of defence. However, wireless IDS, being a new technology, has its drawbacks, too. Then again, with the capability to detect probes, Denial of Service attacks, SSID spoof detection, and a variety of other 802.11 attacks, the benefits of a wireless IDS already outweigh the drawbacks.

Of course, IDS alone is not the solution to any security threat. It is a very small part of a much bigger security infrastructure. WLANs require a number of other security levels to be integrated before an adequate level of security can be achieved. With the current speed of adoption of new technologies, the ever increasing number of potential threats to a system, the growing complexity and sophistication of attacks, as well as attackers... you are doomed to lose your data unless you secure it using IDS, along with other lines of defence. 

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